

DP Computer Science: A 1.3.1–1.3.7 Operating and Control Systems

1. Overview

This section covers the core role of operating systems (OS) in managing computer resources like memory and processors, as well as control systems (HL only). The content is broken down into multiple lessons to allow for hands-on activities and engagement with new knowledge.

2. A1.3.1–A1.3.7 Learning Objectives

Core Objectives

By the end of this section, students will be able to:

- Define an **operating system (OS)** and its core functions
- Explain how an OS manages system resources (**memory, processor, I/O devices**)
- Describe how an OS **abstracts hardware complexities**
- Identify the **different types of OSs** and their characteristics
- Evaluate the importance of an OS in computer systems

IB ATL Skills

Skill	Description
Thinking	Critical thinking (analysing OS functions); Creative thinking (exploring alternative OS designs)
Communication	Explaining OS concepts clearly and concisely
Research	Gathering and evaluating information about different OSs

3. Key Vocabulary

Term	Definition
Operating System (OS)	Software that manages computer hardware and provides a platform for applications
Abstraction	Hiding complex hardware details from users and applications
Process scheduling	The method by which the OS allocates CPU time to processes
Multitasking	Running multiple processes concurrently on a single CPU

Term	Definition
Deadlock	A situation where two or more processes are blocked, each waiting for a resource held by the other
Polling	Continuously checking the status of an I/O device
Interrupt	A signal sent by an I/O device to the CPU to request attention
Control system	An automated system that manages and directs the behaviour of another device (HL only)
Open-loop system	A control system that does not use feedback
Closed-loop system	A control system that uses feedback to continuously adjust

4. Lesson Breakdown

This section contains a lot of content, especially for students new to the topic. It is advised to break it down into **five lessons** and give students hands-on activities throughout.

Lesson 1: Introduction to Operating Systems and Core Functions

Learning Objectives

- Define an operating system and its primary role in a computer system
- Explain how operating systems **abstract hardware complexities**
- Describe core OS functions: **memory management**, **process management**, and **I/O management**

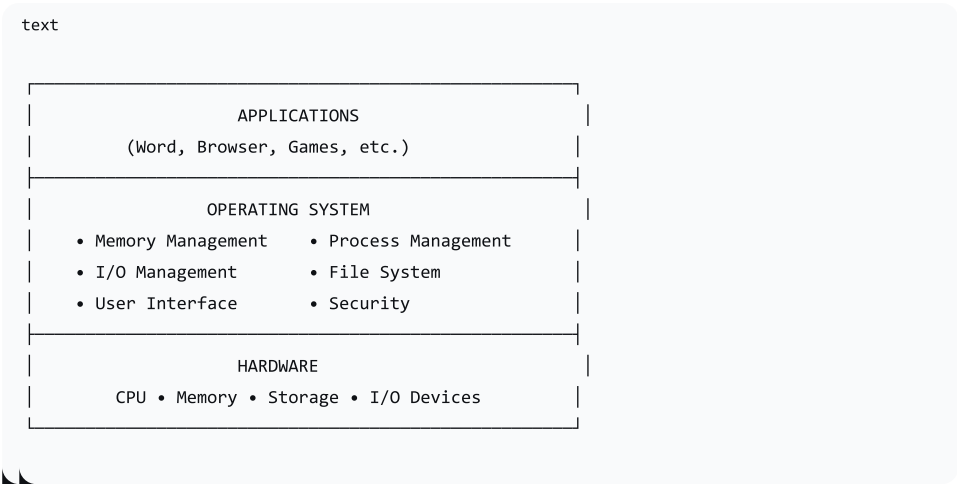
ATL Focus: Thinking (critical thinking, transfer)

Lesson Plan

Phase	Time	Activity
Introduction	10 min	"What makes your computer work?" Brainstorming session
What is an OS?	15 min	Define OS; use analogies (restaurant manager, traffic controller); diagram of OS, hardware, and applications
Abstraction and Resource Management	20 min	Explain abstraction; discuss CPU, memory, storage, I/O management
Activity: OS in Action	15 min	Groups identify OS functions in common tasks (opening a document, browsing the internet)

Core Concepts

What is an Operating System?



Key OS Functions:

Function	Description
Memory Management	Allocates memory to programs; manages virtual memory
Process Management	Schedules CPU time; manages multitasking
I/O Management	Handles input/output devices; abstracts hardware differences
File System	Organises and manages data on storage devices
User Interface	Provides a way for users to interact with the computer
Security	Controls access to resources; protects data

Lesson 2: Process Scheduling and I/O Handling

Learning Objectives

- Explain the concept of **process scheduling** and its importance
- Compare and contrast different scheduling algorithms: **FCFS, round robin, multilevel queue, priority**
- Differentiate between **polling** and **interrupt handling**
- Analyse the factors that influence the choice of I/O handling method

ATL Focus: Research (information gathering) and Thinking (analysis, evaluation)

Lesson Plan

Phase	Time	Activity
Process Scheduling	20 min	Define scheduling; introduce FCFS, round robin, multilevel queue, priority; discuss

Phase	Time	Activity
		advantages/disadvantages
Polling vs. Interrupts	20 min	Explain both methods; discuss trade-offs; real-world examples
Activity: Scheduling and I/O Scenarios	20 min	Students recommend scheduling algorithms and I/O methods for various scenarios

Core Concepts

Scheduling Algorithms:

Algorithm	Description	Advantages
FCFS (First Come, First Served)	Processes executed in order of arrival	Simple, fair
Round Robin	Each process gets a fixed time slice	Fair; good for interactive systems
Priority Scheduling	Higher priority processes run first	Important tasks get priority
Multilevel Queue	Multiple queues with different scheduling policies	Flexible; supports different task types

Polling vs. Interrupts:

Method	Description	When Used
Polling	CPU continuously checks I/O device status	Simple devices; low data rates
Interrupt	Device signals CPU when ready	High-performance systems; keyboard input

Trade-offs:

- Polling: Simple but wastes CPU cycles
- Interrupts: Efficient but more complex to implement

Lesson 3: Multitasking and Resource Allocation (HL Only)

Learning Objectives

- Explain the challenges of **multitasking** and resource allocation in operating systems
- Describe the concept of **deadlock** and its implications
- Analyse strategies for preventing and handling deadlock situations

ATL Focus: Communication (explaining ideas) and Self-Management (reflection)

Lesson Plan

Phase	Time	Activity
Multitasking Challenges	15 min	Context switching overhead, resource contention, synchronisation issues
Deadlock	20 min	Define deadlock; four necessary conditions; real-world examples
Deadlock Prevention and Handling	15 min	Prevention strategies; detection and recovery
Reflection	10 min	Ethical considerations: fairness, security, privacy

Core Concepts

Challenges of Multitasking:

Challenge	Description
Context Switching Overhead	Saving and restoring process states takes CPU time
Resource Contention	Multiple processes compete for limited resources
Synchronisation Issues	Processes may need to coordinate access to shared data

Deadlock — Four Necessary Conditions:

Condition	Description
1. Mutual Exclusion	Resources cannot be shared
2. Hold and Wait	Process holds resources while waiting for others
3. No Preemption	Resources cannot be forcibly taken away
4. Circular Wait	Processes wait in a circular chain

Deadlock Prevention Strategies:

Strategy	Description
Resource Ordering	Processes request resources in a fixed order
Deadlock Avoidance	OS checks if allocation would lead to deadlock
Deadlock Detection	OS periodically checks for deadlock and recovers
Deadlock Recovery	Preempt resources or abort processes

Lesson 4: Control Systems and Their Components (HL Only)

Learning Objectives

- Identify and define the components of a **control system**
- Describe the **input, process, output, and feedback** mechanisms
- Differentiate between **open-loop** and **closed-loop** systems

ATL Focus: Thinking (analysis, evaluation) and Communication (explaining ideas)

Lesson Plan

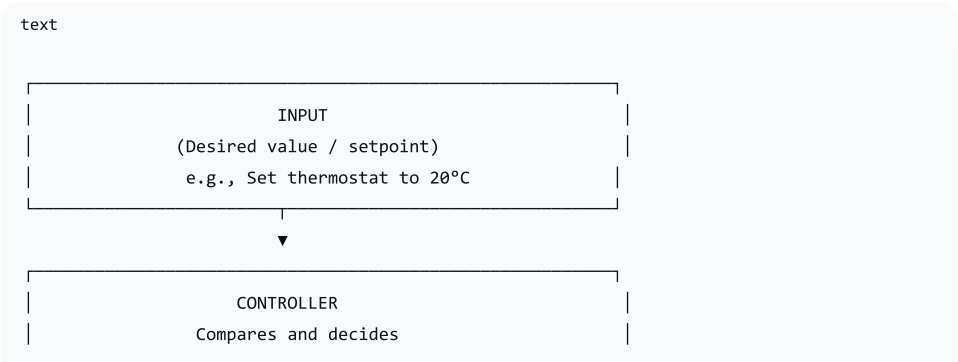
Phase	Time	Activity
Introduction	10 min	Everyday devices that regulate themselves; thermostat analogy
Control System Components	15 min	Controller, sensors, actuators, transducers
Input, Process, Output, and Feedback	15 min	Continuous cycle of a control system
Open-Loop vs. Closed-Loop	10 min	Compare and contrast; examples
Activity: Real-World Scenarios	10 min	Groups classify and identify components of control systems

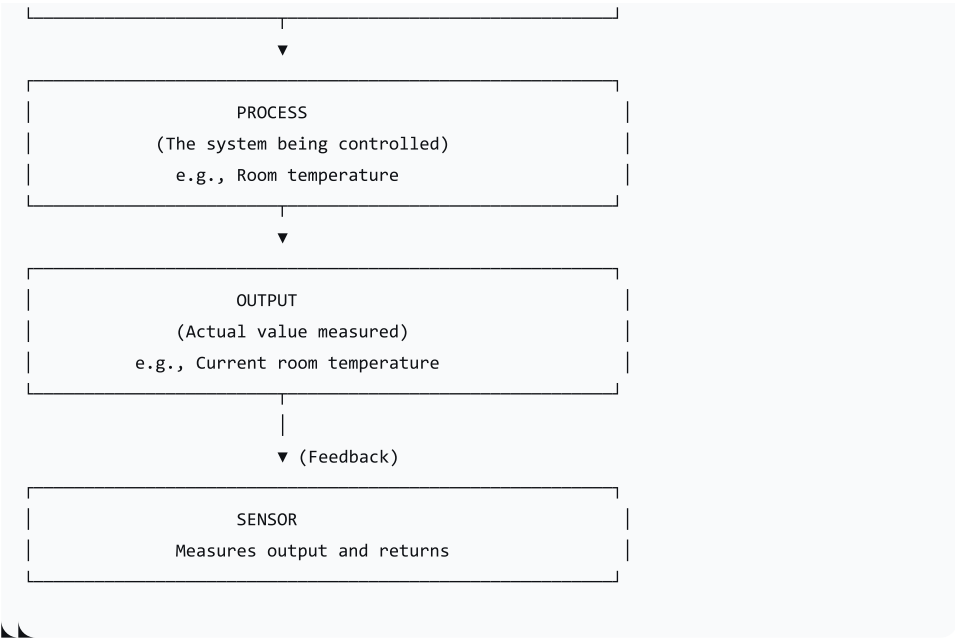
Core Concepts

Components of a Control System:

Component	Role	Example (Thermostat)
Controller	The "brain" that makes decisions	Thermostat
Sensors	The "eyes" that gather information	Temperature sensor
Actuators	The "muscles" that perform actions	Heater/AC
Transducers	Convert one form of energy to another	Thermocouple

Control System Cycle:





Open-Loop vs. Closed-Loop Systems:

Feature	Open-Loop	Closed-Loop
Feedback	No	Yes
Accuracy	Lower	Higher
Stability	Less stable	More stable
Complexity	Simple	Complex
Cost	Lower	Higher
Example	Simple toaster	Thermostat

Lesson 5: Applications of Control Systems (HL Only)

Learning Objectives

- Analyse how control systems are applied in **real-world scenarios**
- Identify the components and mechanisms in various examples

ATL Focus: Thinking (analysis, synthesis) and Communication (presentation)

Lesson Plan

Phase	Time	Activity
Review	5 min	Quick review of control system components
In-Depth Examples	30 min	Autonomous vehicle, traffic signal control, washing machine

Phase	Time	Activity
Group Presentation	20 min	Groups present findings on assigned scenarios; class discussion

Real-World Examples

1. Autonomous Vehicle:

Component	Description
Input	Navigation route, desired speed
Sensors	Lidar, radar, cameras, GPS
Controller	Onboard computer
Process	The vehicle itself
Actuators	Steering, brakes, throttle
Feedback	Continuous sensor data on position, speed, surroundings
Type	Closed-loop

2. Traffic Signal Control System:

Component	Description
Input	Desired traffic flow patterns
Sensors	Traffic detectors (loops, cameras)
Controller	Traffic management computer
Process	Flow of traffic
Actuators	Traffic lights
Feedback	Sensor data on traffic volume to adjust timings
Type	Closed-loop

3. Automatic Washing Machine:

Component	Description
Input	Selected wash cycle and settings
Sensors	Water level sensor, temperature sensor
Controller	Microcontroller
Process	Washing cycle

Component	Description
Actuators	Water pump, heating element, motor
Feedback	Sensor data on water level and temperature
Type	Closed-loop

5. Differentiation

Level	Strategy
Support	Use analogies and visual diagrams; provide step-by-step examples; use interactive simulations for scheduling
Challenge	Research and present on advanced topics (virtualisation, distributed systems, real-time OS); analyse ethical implications

6. Resources

- **Presentation:** 1.3 Operating Systems
- **eBook:** A1.3 Operating and Control Systems
- **Online Resources:** Khan Academy, [Code.org](https://code.org), Crash Course Computer Science
- **Interactive Simulations:** Process scheduling visualisers, control system simulations
- **Real-World Examples:** Embedded systems, real-time systems, mobile operating systems

7. Assessment

Assessment Type	Description
Class Participation	Active participation in discussions and activities
Quizzes	Short quizzes on key concepts
Assignments	Research tasks and problem-solving exercises
Projects (HL)	Research and present on advanced OS topics

8. Summary: Key Takeaways

Topic	Key Takeaway
Operating System	Software that manages hardware and provides a platform for applications

Topic	Key Takeaway
Resource Management	OS manages CPU, memory, I/O, and storage
Abstraction	OS hides hardware complexity from users and programs
Scheduling	Different algorithms suit different system needs
I/O Handling	Interrupts are more efficient than polling for most devices
Deadlock (HL)	Four conditions must exist; can be prevented or recovered from
Control Systems (HL)	Use feedback to maintain desired output; critical in automation